

PramaanAI

Evidence-based document and identity verification for SSB officers at the
India–Nepal and India–Bhutan borders

Smart India Hackathon 2026 · Problem Statement 26188
Ministry of Home Affairs / Sashastra Seema Bal (SSB), Police II Division

Android app · FastAPI server · Web console

Project report · 24 September 2026

Live console: pramaanai-703j.onrender.com · API: bordershield-pramaan-api.onrender.com

Source: github.com/BhartiKumarii/PramaanAI

All registry data in this build is synthetic (mock). No government database is accessed. The system assists the officer; it never declares guilt or denies entry.

1. The problem

SSB guards two open, treaty-based land borders: India-Nepal (1,751 km) and India-Bhutan (699 km). Indian and Nepali citizens cross under treaty right, often with no passport at all, through Border Out Posts and patrols rather than fixed immigration desks. The fraud risk is concentrated in impersonation (someone else's document) and fraudulent or altered documents — not in universal scanning of every crosser.

What makes it hard in the field

- Many document types, three countries: passports, visas and visa stamps, entry permits, driving licences, Aadhaar, citizenship certificates, immigration stamps — in English, Hindi and Nepali (Devanagari) and Dzongkha.
- Unreliable connectivity: a post may be online, on a weak link, or offline at any moment. A tool that stops without a connection is not used.
- No per-post infrastructure: no public evidence of computers or servers at every post — the officer has a phone.
- Checks are manual and inconsistent: MRZ check digits, expiry dates, stamp/checkpoint consistency and photo comparison are slow to do by eye and easy to miss under pressure.
- Decisions must stay with the officer and be explainable afterwards — a bare risk number is not acceptable evidence.
- Privacy: identity documents and faces are sensitive personal data.

2. Proposed solution and how it addresses the problem

One Android app + one FastAPI server + one web console. The officer photographs the document and takes a live photo; the phone does the fast, private work (finding regions, reading text, validating the MRZ, liveness); the server runs the heavy checks and returns an explainable result; the officer decides — Clear, or Send to admin — and an admin answers from the web console.

Problem	How PramaanAI addresses it
Impersonation	Live photo compared with the document photo (InsightFace); liveness prompts (blink, head turn) and an on-device anti-spoof model; identity graph shows the same face under another name, or the same document number under another name.
Altered / fraudulent documents	MRZ check digits (ICAO 9303) and MRZ-vs-printed consistency; QR / digital-signature checks; region-level image forensics; stamp checked against the official checkpoint reference; date logic; registry comparison (mock).
Many document types and scripts	Region detection (YOLO11n) + PP-OCR for Latin text and a Devanagari recogniser for Nepali/Hindi; document type from MRZ, keywords and regions; e-Visa, e-Aadhaar, DigiLocker and online permits recognised; PDFs accepted.
Connectivity	Live Online / Weak / Offline check; offline captures are encrypted (Android Keystore) and verified automatically when the link returns; retries are idempotent.
Explainability	Every result names the check and the reason, marks the exact place on the document, and writes a suggested reason for the officer's decision.
Officer decides	The system never denies entry or declares guilt: Clear or Send to admin; the admin sees a suggested action and confirms it.
Privacy	Only region crops and phone-read text are sent for verification — never the full frame; evidence images are attached to a case only for admin review; Aadhaar numbers masked; raw OCR text not stored.

3. Innovation and uniqueness

- Border-specific rules engine: India–Nepal treaty nationals need no visa or stamp; India–Bhutan post-2022 rules; designated crossings for third-country nationals — the result is judged against the crossing, not a generic checklist. The crossing comes from the stamp, the officer's post, or the officer.
- Explainable, evidence-first results: named reasons, problem boxes on the original document, field sources (server OCR / MRZ / QR / phone / Devanagari), and a risk breakdown — never a mystery score.
- Split intelligence for weak links: a 10.6 MB YOLO11n model on the phone sends only crops in one request; everything works offline and syncs later.
- Two-sided liveness: active prompts (blink, head turn — the same tracked face must complete both) plus a passive anti-spoof model, both on the phone; the server reports them as evidence.
- Devanagari document reading with Nepali dates kept in Bikram Sambat — never mis-read as Gregorian.
- Closed loop with the admin: Send to admin carries the evidence; the admin sees an automatically suggested action (Clear / Re-capture / Manual review) and the answer returns to the officer's phone.
- Honest by design: mock data is labelled everywhere; checks that cannot run say so (e.g. e-Visa must be confirmed on the official portal) instead of faking a confident answer.

4. Technical approach

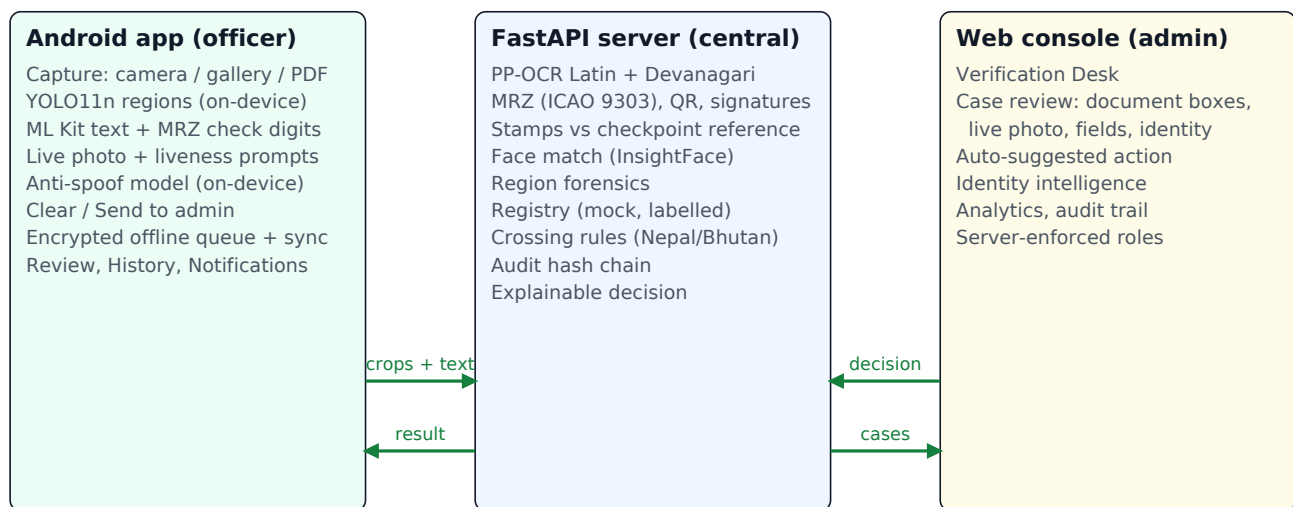


Figure 1 — Architecture: the phone sends crops and text; the server returns an explainable result; the web console answers cases.

Technology stack

Layer	Technologies
Android app	Kotlin, Jetpack Compose, Material 3, CameraX, ML Kit (text, face, barcode), ONNX Runtime (YOLO11n regions, MiniFASNet-V2 anti-spoof), PdfRenderer, Retrofit, WorkManager, Android Keystore (AES-GCM), 7 UI languages
Server	Python, FastAPI, Pydantic, SQLAlchemy, Alembic, PostgreSQL (SQLite for local dev), JWT + Argon2, RapidOCR / PP-OCR (ONNX) incl. PP-OCRv5 Devanagari, InsightFace (SCRFD + ArcFace-family), OpenCV, cryptography (signatures), NetworkX (identity graph)
Web console	React 19, TypeScript, Vite, Tailwind CSS, Recharts
Operations	Docker, Render (API + PostgreSQL + static site), GitHub Releases (APK)

Verification checks (server)

Area	Checks
Machine-readable	MRZ structure and check digits (TD1/TD2/TD3, MRV-A/B); MRZ vs printed details; QR/barcode decoding and QR vs printed; digital signature against a configured trust store
Content	Key fields read; date order and validity; number formats; nationality codes; cross-document consistency (passport vs visa)
Stamps and rules	Stamp detection and text reading; checkpoint matched to the official reference; stamp vs visa; India-Nepal / India-Bhutan crossing rules
Face and liveness	Document photo quality; secondary (ghost) portrait; face match with the live photo; liveness report from the phone
Image integrity	Region-vs-surroundings forensics (noise, sharpness, error level, JPEG grid, duplicates); capture quality
Registry and identity	Mock registry lookup (labelled); duplicate document number; face cluster (same face, other names)

5. Methodology and workflow

The design follows four rules taken from the problem: work at the post with a phone, never block on connectivity, explain every flag, and leave the decision to the officer. Each check is a separate, swappable service (OCR, MRZ, face, registry, forensics) with a clearly labelled mock where a real government system would sit.

Step	What happens
1 Capture	Officer photographs the document (or picks a photo / PDF).
2 On the phone	YOLO11n finds photo, MRZ, QR, stamps; ML Kit reads text; MRZ check digits verified.
3 Review + crossing	Officer sees the regions and text; crossing comes from the post or is chosen.
4 Live photo	Blink + head-turn prompts, anti-spoof score; or skip.
5 Connectivity	Live health check: Online / Weak / Offline. Offline = encrypted queue, auto-sync.
6 Server checks	OCR, MRZ, QR/signature, stamps, face, forensics, registry (mock), crossing rules.
7 Explainable result	Status + named reasons + problem boxes on the document + identity links.
8 Officer decides	Clear, or Send to admin with an auto-written, editable reason.
9 Admin responds	Web console: evidence, suggested action; the answer returns to the phone.
10 Audit	Every step is logged in a tamper-evident hash chain.

Result states

Verified (all blocking checks pass) · Review required (a named inconsistency) · Not verified (a check could not run, e.g. an unreadable region) · Official verification required (e.g. Aadhaar or e-Visa, which only the issuing service can confirm). Advisory checks (liveness, Devanagari text, electronic documents) are shown but never change the status on their own.

6. Implementation process

Phase	Delivered
1 — Server	FastAPI services, models and migrations, mock registries, explainable decision engine, audit hash chain; verified through /docs and automated tests.
2 — Android	On-device region detection and text/MRZ, live connectivity state, encrypted offline queue with sync, Verify-document flow, liveness, PDF input.
3 — Web console	Verification Desk, case review with document evidence, automatic suggested action, identity intelligence, analytics, audit trail; public landing page.
4 — Integration	End-to-end wiring check (phone → server → admin → phone) run against the live deployment; UI polish and translations.

Demo posts and accounts

The demo deployment is set on SSB's own borders. Each post matches the official checkpoint reference data, so an officer's assigned post fills in the crossing and its rules automatically.

Post (code)	Border	Location	Demo login
Raxaul (RAX)	India-Nepal	Bihar	raxaul_officer
Sunauli (SUN)	India-Nepal	Uttar Pradesh	sunauli_officer
Jaigaon (JGN)	India-Bhutan	West Bengal	jaigaon_officer
Web console (admin)	—	—	it_admin

Password for all demo accounts: BorderShield123. All case data in the demo is synthetic.

Quality evidence

- 187 automated server tests pass (unit and API).

- End-to-end wiring check: 24/24 on the live Render deployment (verify → case → evidence → send to admin → admin decision → the phone sees the answer → audit chain intact).
- On-device parity test for the anti-spoof model: Android scores match the Python reference within 0.05.
- Fresh-database migrations verified on SQLite and PostgreSQL 16.

7. Measured results

Synthetic documents were generated by the project with layouts matching its own templates, so synthetic numbers are optimistic and not evidence of real-world accuracy. The real set is 57 labelled photographs of real documents (kept local, never published).

Measure	Result
Overall status matches expectation (46 synthetic cases)	0.891
Expected check-level outcomes (synthetic)	53 / 56
Needs-attention detection, F1 (synthetic)	0.925 (precision 0.939, recall 0.912)
OCR key fields (synthetic)	18 / 18
Document type, real photographs	0.877 (57 documents)
Phone path vs server path (39 real captures replayed)	status 38/39, type 39/39
Devanagari text on Nepal ID / citizenship / permit / Aadhaar	0 characters before; now read (conf. 0.76–0.93), +0.07 s/doc
Speed-ups (6-image profile)	67.8 s → 45.6 s, identical accuracy
Server memory (3 + 2 parallel verifications)	685 MB → about 420 MB peak
Tampering detection (synthetic)	recall 0.33 — the weakest module (see §8)

Where it is weak — stated plainly

- Pixel-level tampering (a re-compressed photo or pasted text) is often missed; flags are advisory.
- Handwritten entries on visa stamps are not read reliably: 0/7 with the current OCR, 1/7 even on hand-placed crops. The officer reads them from the document image.
- Anti-spoof was checked on a handful of images: printed/document faces were flagged, but a clean digital passport photo passed as real — liveness prompts are the stronger signal; the score is advisory until calibrated on field captures.
- Bengali, Gurmukhi and Dzongkha document text is not read; the Dzongkha UI is partly English.

8. Feasibility and viability

Aspect	Assessment
Technical	Runs today on a mid-range Android phone and a 512 MB cloud instance; all models are open (Apache-2.0 / MIT-compatible) and run on CPU with ONNX Runtime.
Operational	Fits existing SSB practice: the officer keeps the decision; no new hardware at posts; works offline; 7 UI languages.
Integration	Every external dependency (registry, UIDAI, e-Visa status, DigiLocker, ePassport PKI) sits behind an interface with a labelled mock, ready for an authorised API.
Cost	Open-source stack; one central server scales horizontally (stateless API, bounded concurrency, 503 + retry when saturated).
Security and privacy	See section 9.

9. Security

Security is applied at every layer: the officer's phone, the network, the server and the stored records. Every item below is implemented in this build.

Layer	Measure	How
Access	Sign-in	JWT access tokens (30 min) and refresh tokens (24 h); passwords hashed with Argon2; login limited to 5 attempts per minute per address.
Access	Roles	Enforced on the server, not only in the UI: officers verify and send; only admins (Reviewer) decide cases, manage users and devices, or remove data.
Access	Session	Auto-lock after inactivity with a warning; tokens kept in encrypted storage on the phone; lost devices can be disabled or revoked by the admin.
Transport	Encryption in transit	HTTPS (TLS) between the phone, the web console and the server.
Phone	Encryption at rest	Captures and the offline queue are encrypted with AES-GCM using a key held in the Android Keystore; images are deleted automatically after the retention period.
Data	Minimisation	Only detected regions and phone-read text are sent for verification — never the full frame; the verify endpoints process images in memory and keep only SHA-256 hashes; raw OCR text is not stored; Aadhaar is masked (XXXX XXXX 1234); evidence images are attached only when a case is opened for review.
Data	Input limits	Uploads ≤ 10 MB and ≤ 40 MP, 1–4 images; region crops covering $> 90\%$ of the frame are rejected; every request is schema-validated.
Integrity	Tamper-evident records	Each verification is hashed (SHA-256), linked to the previous record and signed with HMAC-SHA256; the whole chain can be re-verified at any time.
Integrity	Audit trail	Created, viewed, sent, decided and removed events are logged with the actor; removed test data stays in the chain, marked withdrawn.
Integrity	Document signatures	Signed QR data is checked against a configured trust store (Ed25519); altered data fails the check.
Identity	Anti-impersonation	Face match with the live photo, blink / head-turn liveness prompts and an on-device anti-spoof score.
Server	Hardening	Secrets only in environment variables; personal data redacted from server logs; bounded concurrency (503 + retry when busy).
Ethics	Fairness	No emotion, skin temperature, nationality, religion or ethnicity used as a signal; wording never accuses; mock data labelled everywhere.

Security workflow — one verification, end to end

Step	What protects it
1 Sign in	Argon2-checked password, rate-limited login, short-lived JWT; role attached to the token.
2 Capture	Image stays on the phone, encrypted (AES-GCM, Keystore key).
3 Minimise	Phone finds the regions; only crops and text are prepared — never the full frame.
4 Send	TLS; bearer token checked; offline requests wait encrypted and are retried safely (same request id = same record, never a duplicate).
5 Check	Server validates size and schema, processes images in memory, stores only hashes and a minimised result.
6 Record	Result hashed, linked to the previous record, HMAC-signed; audit event written.
7 Decide	Officer clears or sends to admin; only an admin (server-checked role) can decide the case; every decision is audit-logged.
8 Retain	Phone images auto-deleted after the retention period; session auto-locks; the chain can be re-verified at any time.

10. Challenges and the strategy used to solve them

Challenge	Strategy
Weak or no connectivity	Phone-side detection and MRZ; one small request of crops; live Online/Weak/Offline check; encrypted queue; idempotent sync.
Phone APK size (227 MB)	ARM-only native libraries and R8 shrinking → 97 MB, with ML Kit keep rules to avoid release-only crashes.
Same person reported as 'no match'	Tight crops had no detectable face; added padded detection variants, 'no face' = inconclusive (never a mismatch), and a 'possible match → review' band.
Two portraits on passports/licences flagged	Recognised the secondary (ghost) portrait as a normal feature and compared it with the main photo instead.
Nepali/Hindi documents unreadable	Devanagari recogniser on the boxes the Latin pass drops; separate fields; Bikram Sambat kept.
Handwriting	Measured three approaches; none reliable → not shipped, documented honestly.
Server out of memory on Render	Profiled per model; MediaPipe loaded only when used; shared OCR engine pool; one verification at a time; malloc tuning → 685 MB to about 420 MB.
Photos or screens held up to the camera	Random blink / head-turn prompts on the same tracked face + on-device anti-spoof score.
Explainability for the admin	Problem boxes on the document, identity links, and an automatic suggested action with its reasons.
Database differences (SQLite vs PostgreSQL)	Found write-order and ID-type bugs by running the full flow against PostgreSQL 16; fixed migrations for fresh databases.

11. Impact and benefits

- Faster, more consistent checks: MRZ digits, dates, stamps and face comparison in one flow instead of by eye — the officer spends time on the cases that need it.
- Fewer impersonation misses: live photo + liveness + identity graph target the main real-world risk.
- Works where the border is: offline-first, phone-only, in the officer's language.
- Accountability: every decision is explained, reviewed when needed, and recorded in a tamper-evident audit trail.
- Fair by design: no emotion, skin temperature, nationality, religion or ethnicity used as a signal; the wording never accuses; the officer decides.
- Ready to connect: swappable services let authorised government APIs replace the mocks without changing the app.

12. Screens (synthetic data)

PramaanAI

Overview

Verification Desk

5

Document Verification

Identity Intelligence

Checkpoints

Area Monitoring

Officers Monitoring

Device Management

Users

Analytics & Intelligence

Audit Trail

Admin Settings

Verification Desk

Officer it_admin Log out

Verification Desk

Cases sent from the app's Verify-document flow, with their verification result — your operational workspace.

☐ Include 12 cases from the earlier screening flow and demo data (no document-verification record)

AWAITING REVIEW

2

HIGH PRIORITY

0

RE-CAPTURE REQUIRED

1

MANUAL REVIEW

0

OVERDUE (>24h)

0

Q Search case, name, checkpoint...

All Statuses

All

High

Medium

Low

All Checkpoints

All Documents

Priority

Longest Waiting

Newest First

CASE ID	PERSON	DOCUMENT	CHECKPOINT	PRIORITY	STATUS / RISK	WAITING	ASSIGNED	ACTION
BSA-20260924-0003	ANANYA SYNTHETIC S... INDIAN	Indian Passport India	RAX	Medium	Re-capture Required Review required Medium Risk (36/100)	6h	Unassigned	Review
BSA-20260924-0002	ANANYASYNTHETIC SH... INDIAN	Indian Passport India	RAX	Medium	Submitted for Review Review required Medium Risk (35/100)	6h	Unassigned	Review
BSA-20260924-0001	ANANYASYNTHETIC SH... INDIAN	Indian Passport India	RAX	Medium	Submitted for Review Review required Medium Risk (30/100)	6h	Unassigned	Review

Showing 3 of 3 cases (12 earlier hidden)

Refresh

Verification Desk — cases from the app with their verification result; earlier-flow and demo cases behind a toggle.

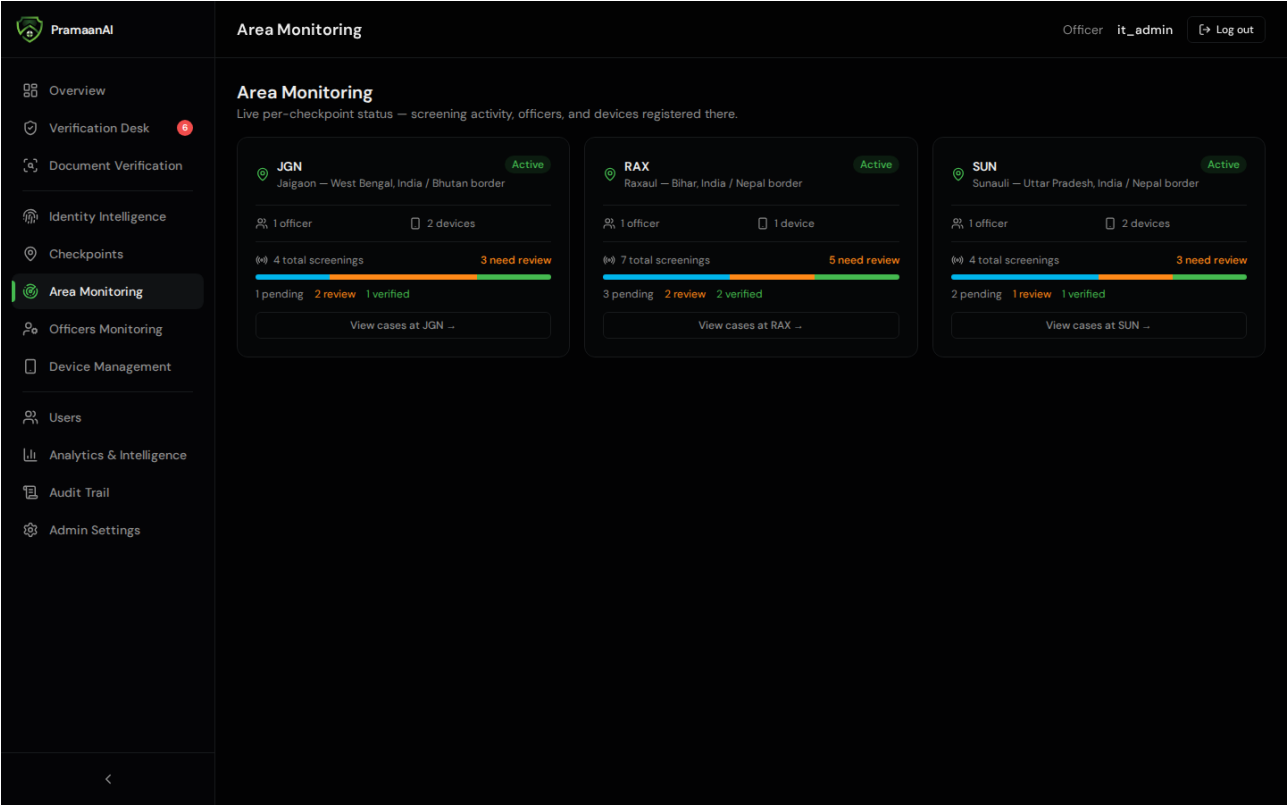
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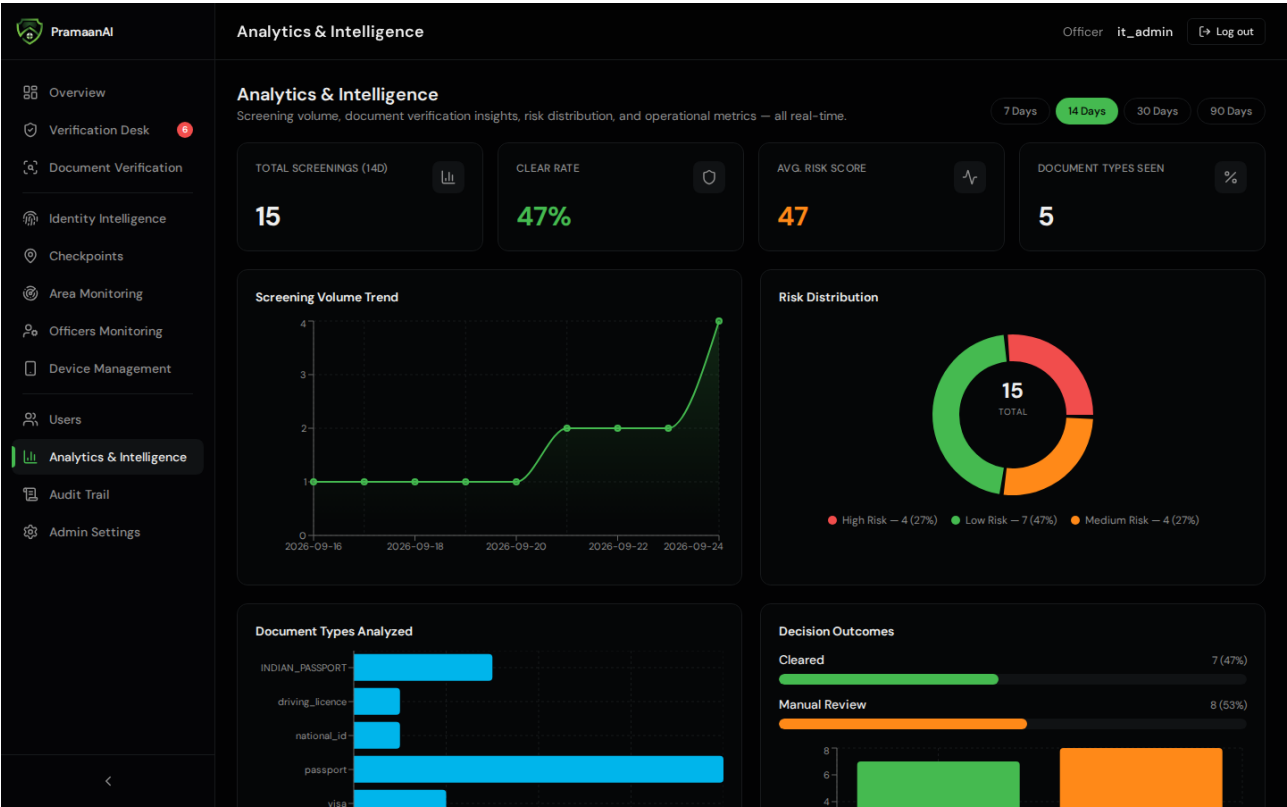
Case review — document with problem areas, face match, identity links, fields with sources, suggested wording.

Take action — automatic suggestion written from the checks; 'Use suggestion' pre-fills the form, the admin confirms.





Area Monitoring — per-post activity at Jaigaon, Raxaul and Sunauli.



Analytics & Intelligence.

13. Future work

Idea	Implementation and methodology
Officer assistant chatbot	A chat panel in the app and console that answers procedure questions ("Does a US national need a visa at Sunauli?", "Which crossings are designated for foreigners?") in the officer's language. Retrieval-augmented generation over the project's own reference data (border rules, checkpoint list, SOPs) with citations; server-side LLM with an offline FAQ fallback on the phone. It explains rules and results — it never makes the decision.
Geo-fencing	Checkpoint polygons in the reference data; GNSS/NavIC location on the phone fills the post automatically and tags each verification; advisory when outside a designated crossing; mock-location and impossible-speed checks.
CCTV at crossings	Edge box reads RTSP feeds: YOLO person/vehicle detection and tracking, line-crossing counts, loitering and night-movement alerts, number-plate reading against the (mock) vehicle-permit registry. No blanket face recognition — only with legal authorisation (DPDP Act 2023).
Satellite	Satellite messaging for posts with no network (tiny case summaries, full sync later); Sentinel-2 / Bhuvan change detection along the border to suggest patrol areas; NavIC for location.
ePassport chip (NFC)	Read DG1/DG2 with BAC/PACE keyed from the MRZ; compare chip data and photo with the printed page and the live photo; passive authentication once an authorised CSCA trust list is available.
Official e-document checks	Authorised APIs for e-Visa status, UIDAI secure QR, DigiLocker / Parivahan in place of today's 'confirm on the official service' guidance.
Handwriting	Label handwritten-field boxes (new YOLO class) and fine-tune a handwriting recogniser on border forms.
More scripts	Bengali, Gurmukhi and Dzongkha document text; complete Dzongkha UI with a native reviewer.
Calibration	Field pilot to calibrate the anti-spoof threshold and forensic flags on real captures before any check is made blocking.

Planned security

Addition	Purpose
Certificate pinning + mutual TLS	The app talks only to the genuine server; only registered devices can connect.
Device attestation (Play Integrity)	Refuse rooted, emulated or tampered phones.
Multi-factor sign-in for admins	Hardware key or OTP for the web console.
Keys in a KMS / HSM with rotation	Signing and encryption keys never leave secure hardware.
Encryption of stored records	Field-level encryption of personal data in the database.
External anchoring of the audit chain	Periodic signed timestamps of the chain head from a trusted time-stamping authority.
Post-scoped access	Officers see only their own post's data; supervisors their area.
Security monitoring	Alerts on unusual sign-ins, bulk exports or repeated failed checks.
Shared rate limiting	Login and API limits enforced across all server instances.
DPDP Act 2023 compliance	Retention schedules, purpose limitation and data-subject procedures agreed with MHA / SSB.
Independent security audit	VAPT by a CERT-In empanelled auditor before field deployment.

Data notice: registry lookups in this build use fictional mock data only; no real government database is accessed. Real document photographs used for evaluation are kept locally and are not published.